Sharif University of Technology
Department of Physics

QFT 1: Problem Set 1

Professor: Arash Arabi Ardehali

Student: Abolfazl Balouch

E-mail: Balouch1381@gmail.com

October 20, 2026

Problem

Consider a 4-dimensional hyper-cubic lattice with the lattice spacing Δ. Compute the "lattice propagator" as the sum over all lattice paths connecting the lattice points 0 and x,

$$\sum_{\text{lattice path } 0\to x} e^{-m_0 L(\text{path})},$$

where $L(\text{path})$ is the number of links in the path. The path may go through a given link any number of times.

Consider the limit $\Delta \to 0$. Observe that at fixed m_0 the sum either diverges, or is dominated by the shortest path from 0 to x. Determine how $m_0 = m_0(\Delta)$ should depend on Δ for long paths to remain relevant in the limit $\Delta \to 0$.

Suggestion: follow the steps outlined in Lecture 7, with suitable modifications to take into account the lattice structure. The representation

$$\delta_{n,0} = \int_{-\pi}^{\pi} \frac{dk}{2\pi} e^{ikn}$$

of the Kronecker delta may be useful.

Solution

Suppose point x is on the lattice and $x = \Delta N$ ($N = (N_1, ..., N_4)$ is the integer coordinates of the lattice. meaning that $N_\mu = \frac{x_\mu}{\Delta}$.) If $C_\Delta(N)$ is the number of n-step paths from 0 to N, we have:

$$G_\Delta(N) = \sum_{n=0}^{\infty} e^{-m_0 n} C_n(N). \tag{1}$$

In four dimensions, $\delta_{n,0} = \int_{-\pi}^{\pi} \frac{dk}{2\pi} e^{ikn}$ takes the form $\delta_{N,0} = \int_{-\pi}^{\pi} \frac{d^4k}{(2\pi)^4} e^{ik\cdot N}$. In one step, we can move in both the $\hat{e}_\mu$-direction and the $-\hat{e}_\mu$-direction, so the coefficient corresponding to one step in Fourier space is equal to $e^{ik_\mu} + e^{-ik_\mu}$. The sum over the four directions is equal to

$$\sum_{\mu=1}^{4} \left(e^{ik\mu} + e^{-ik\mu}\right) = 2\sum_{\mu=1}^{4} \cos k_\mu \tag{2}$$

Therefore, for n steps we have:

$$C_n(N) = \int_{-\pi}^{\pi} \frac{d^4k}{(2\pi)^4}\, e^{-ik\cdot N} \left(2\sum_{\mu=1}^{4} \cos k_\mu\right)^n \tag{3}$$

Substituting Equation (3) into Equation (1), we obtain:

$$G_\Delta(N) = \int_{-\pi}^{\pi} \frac{d^4k}{(2\pi)^4}\, e^{-ik\cdot N} \sum_{n=0}^{\infty} \left[e^{-m_0}\, 2 \sum_{\mu=1}^{4} \cos k_\mu \right]^n \tag{4}$$

We use the geometric series.($\sum_{n=0}^{\infty} r^n = \frac{1}{1-r}$)

$$G_\Delta(N) = \int_{-\pi}^{\pi} \frac{d^4k}{(2\pi)^4}\, \frac{e^{-ik\cdot N}}{1 - 2e^{-m_0} \sum_{\mu=1}^{4} \cos k_\mu} \tag{5}$$

To see continuum physics, we need to look at small momenta ($k_\mu \ll 1$ because $p_\mu = \frac{k_\mu}{\Delta}$ and p_μ is finite therefor when $\Delta \to 0 \Rightarrow k_\mu \to 0$):

$$\cos k_\mu = 1 - \frac{k_\mu^2}{2} + \frac{k_\mu^4}{24} + \cdots \Rightarrow 2\cos k_\mu = 2 - k_\mu^2 + O(k_\mu^4) \tag{6}$$

Summing over the four dimensions:

$$2\sum_{\mu=1}^{4} \cos k_\mu = 8 - k^2 + O(k^4) \quad , (k^2 = \sum_{\mu=1}^{4} k_\mu^2) \tag{7}$$

So the denominator of Equation (5) becomes as follows:

$$1 - 8e^{-m_0} + e^{-m_0} k^2 + O(k^4) \tag{8}$$

We need to bring the denominator into the familiar form $\frac{1}{p^2+m^2}$. ($p_\mu = \frac{k_\mu}{\Delta} \Rightarrow k_\mu = \Delta p_\mu \Rightarrow k^2 = \Delta^2 p^2$) Also, from Equation (8) we see that the coefficient of p^2 must be $e^{-m_0}\Delta$; therefore:

$$\begin{aligned} &1 - 8e^{-m_0} + e^{-m_0}\Delta^2 p^2 = e^{-m_0}\Delta^2(p^2 + m^2) \Rightarrow 1 - 8e^{-m_0} = e^{-m_0} m^2 \Delta^2 \\ &\Rightarrow e^{m_0} - 8 = m^2\Delta^2 \Rightarrow e^{m_0} = 8 + m^2\Delta^2 \Rightarrow \boxed{m_0(\Delta) = \ln(8 + m^2\Delta^2).} \end{aligned} \tag{9}$$

We write Equation (9) in the following form:

$$m_0 = \ln 8 + \ln\left(1 + \frac{m^2\Delta^2}{8}\right) \tag{10}$$

For $\Delta \to 0$ Using $\ln(1+\epsilon) = \epsilon - \frac{\epsilon^2}{2} + \cdots$, we thus obtain:

$$\boxed{\boxed{m_0(\Delta) = \ln 8 + \frac{m^2}{8}\Delta^2 + O(\Delta^4)}} \tag{11}$$

AI use declaration

Model(s) used:

ChatGPT (GPT-5.6)

Revisions/corrections made to model output, and clarifications added:

I didn't give the full question to the AI, but I used it in the solution process. For writing the LaTeX code, the parts I used from it in the conceptual process of the question are $K_\mu = p_\mu/\Delta$ and generally concepts related to lattices.

Comparisons/checks against the lecture notes:

Before reading the question, I thoroughly reviewed the lecture notes, performed all intermediate calculations, and wrote the question based on that.

Calculation steps, limiting cases, or consistency checks verified by hand:

Since this problem did not involve complex calculations, I handled all the calculations myself.

Calculation steps, limiting cases, or consistency checks subjected to automated verification (via Mathematica, python, LLMs):

As I said above I didn't give the full question to the AI, but I used it in the solution process. For writing the LaTeX code, the parts I used from it in the conceptual process of the question are $K_\mu = p_\mu/\Delta$ and generally concepts related to lattices.